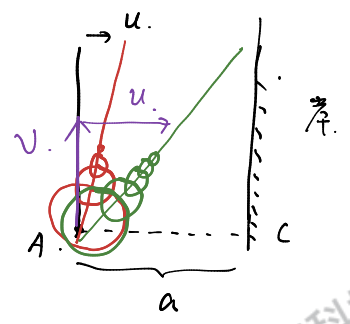
35套.

题1.

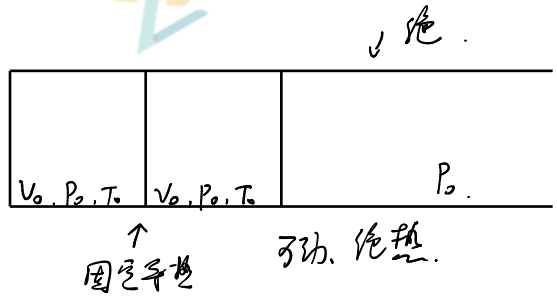


纲. 水中: $v > c$.

$u \le c$. $\Rightarrow$ $x = 0$.

$u > c$. $\Rightarrow$ $x = \frac{a}{u} \cdot v$.

题2.



$C_v = \frac{5}{2}R$.

(1). 等压. $\Rightarrow$ $\begin{cases} \nu C_p \cdot \Delta T = Q_0 \\ \nu C_v \cdot \Delta T = \Delta U \end{cases}$ $\Rightarrow \Delta U = \frac{\frac{5}{2}R}{\frac{7}{2}R} \cdot Q_0 = \frac{5}{7}Q_0$

(2) 设从 $2V_0$ 开始台等. 设右器气体末变大 ΔV_2.

左: $2\Delta U_2 = Q_0 - P_0 \cdot \Delta V_2$.

右: 等压. $P_0 \cdot \Delta V_2 = \nu R \cdot \Delta T_2$.

$\Delta U_2 = \nu C_v \cdot \Delta T_2$.

$\Rightarrow \Delta U_2 = \frac{5}{12}Q_0$.

(3). $Q_{导} = \frac{1}{2} \times \frac{5}{12}Q_0 = \frac{5}{24}Q_0$

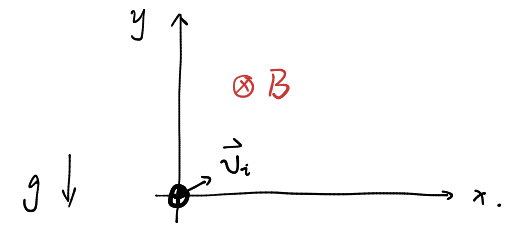
左升温: $\nu C_v \cdot \Delta T_左 = Q_{导} = \frac{5}{24}Q_0$

右升温: $\nu \cdot C_p \cdot \Delta T_右 = \frac{19}{24}Q_0$.

$\nu R T_0 = P_0 V_0$ $\Rightarrow$ ν. ↑

$\Rightarrow \Delta T = \Delta T_右 - \Delta T_左 = \cdots = \frac{Q_0}{7P_0V_0} \cdot T_0$.

题3.



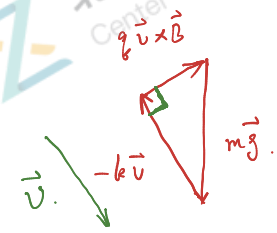
m, q.

$\vec{f} = -k\vec{v}$.

法一: 复: $\hat{x}$: $\hat{y}$: $\Rightarrow$ 降阶分.

法二: 决: $\tilde{z} = x + iy$. $\Rightarrow$ $\tilde{v} = \dot{\tilde{z}}$ $\dot{a} = \dot{v}$.

(1).



等速: 力平衡.

$(kv)^2 + (qvB)^2 = (mg)^2$.

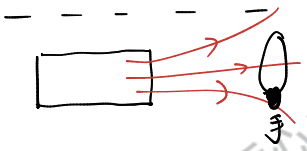
$\Rightarrow v = \frac{mg}{\sqrt{k^2 + q^2B^2}}$

(2). $V_i = 0$.

① 不考虑相对论: $v_{任意时刻} \ll c$.

② 也不能随便换参. $\Rightarrow$ $v \ll c$ 时.

电磁场变换. 有可观值



条形磁铁手中: 动生. $q\vec{v}\times\vec{B}$.

手手中, 感生: $\vec{E}_{涡}$.

惯性. 牛二: $m \cdot \frac{d\vec{v}}{dt} = q\vec{v}\times\vec{B} - k\vec{v} + m\vec{g}$.

合 $\vec{v}_0$ 参中. 物体速度为: $\vec{v}'$. 则 $\vec{v} = \vec{v}_0 + \vec{v}'$.

地参牛二: $m\frac{d(\vec{v}_0+\vec{v}')}{dt} = q(\vec{v}_0+\vec{v}')\times\vec{B} - k(\vec{v}_0+\vec{v}') + m\vec{g}$

$\Rightarrow m\frac{d\vec{v}'}{dt} = q\vec{v}'\times\vec{B} - k\vec{v}' + \underbrace{\left(q\vec{v}_0\times\vec{B} - k\vec{v}_0 + m\vec{g}\right)}_{\text{↑ 未必平衡}=0}$

$\Rightarrow \int m\,d\vec{v}' = \int q\,d\vec{r}'\times\vec{B} - \int k\,d\vec{r}'$

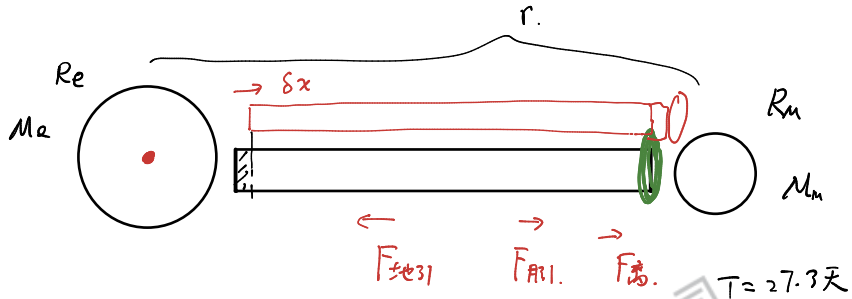
$\Rightarrow m(0 - (-\vec{v}_0)) = q\cdot\vec{S}'\times\vec{B} - k\cdot\vec{S}'$



$\Rightarrow S = \frac{mv_0}{\sqrt{k^2 + q^2B^2}}$

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题4.



T=27.3天

(1). 法一. 合力求功. $\vec{F}_{地引} = \vec{F}_{月引} + \vec{F}_{离}$ ⇒ ✓

法二：虚功：① $S\cdot\delta x$：从地表到月表：

$\delta E_{p地} + \delta E_{p月} + \delta E_{p离}$

$E_{p离}$：离心势：$E_{p离} = -\int \Delta m\cdot\omega^2\cdot r\cdot dr = -\frac{1}{2}\Delta m\cdot\omega^2\cdot r^2$

② M_0：移了 δx.

引力对它做功：$\vec{F}_{地}\cdot\delta\vec{x} + \vec{F}_{月}\cdot\delta\vec{x}$

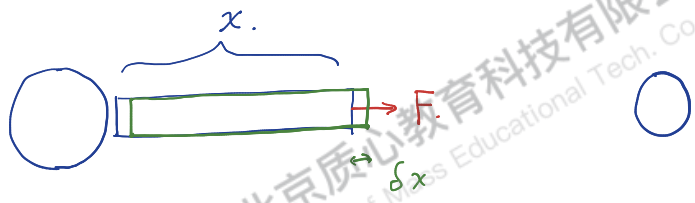
⇩负功　⇧正功.

离心力：$M_0\cdot\omega^2\cdot r\cdot\delta x$.

虚功：$\delta W_{合} = 0$ ⇒ $M_0 = \cdots$ ✓.

$M_0 = 3.60\times10^8 kg$.

(2)



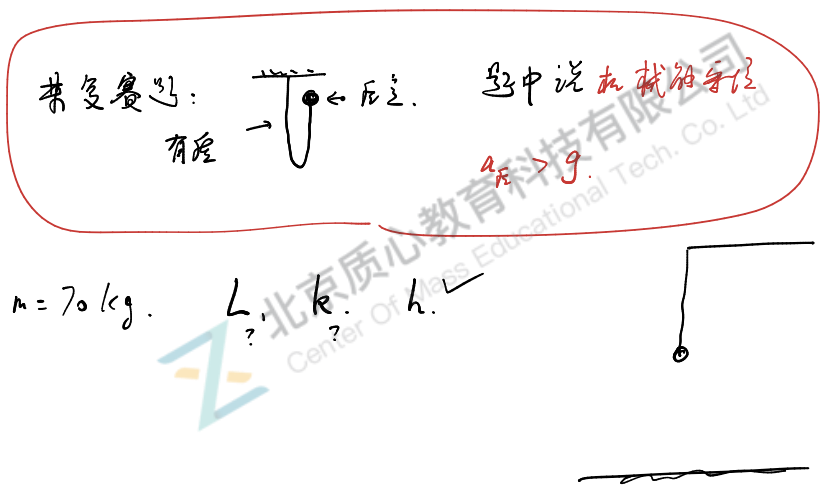
对象：地表到 x 一段天梯：质位移 δx.

$S\cdot\delta x$. 从地表到 x 处：$E_{p地}$. $E_{月}$. $E_{p离}$

$\delta W_{外} = F\cdot\delta x$. ⇒ ..

$\begin{cases} r_L = 3.26\times10^8 m. \\ P_L = 5.83\times10^8 Pa. \end{cases}$

题5.



(1). 能量：水面速为 0：

$mgh = \frac{1}{2}k\cdot(h-L)^2$. ①.

水面.牛二：$ma_0 = k(h-L) - mg$. ②

⇒ $L = 30m$. $k = 35N/m$.

(2) $mg = k\cdot\Delta x_1$ ⇒ $\Delta x_2 = 20m$

(3) V_{max}. 平衡注意处.

⇒ 能：$mg\cdot(L+\Delta x_2) = \frac{1}{2}mV_{max}^2 + \frac{1}{2}k\cdot\Delta x_1^2$.

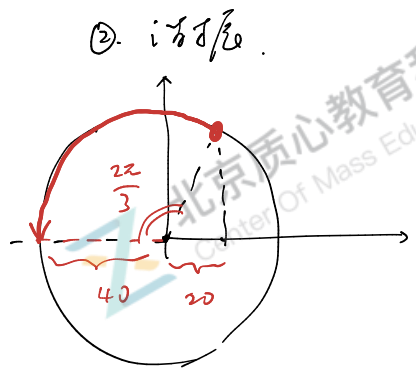
⇒ $V_{max} = 28.3 m/s$.

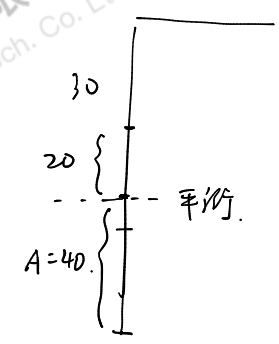
(4) $\omega = \sqrt{\frac{k}{m}} = 0.71\cdot rad/s$.

$A = h - (L+\Delta x_2) = 40m$.

(5) ①. 自由落体：$t_1 = \sqrt{\frac{2L}{g}}$

②. 谐振.





⇒ $t_2 = \frac{T}{3} = \frac{2\pi}{3}\cdot\sqrt{\frac{m}{k}}$

⇒ $t = t_1 + t_2 = \cdots = 5.41 s$.

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题6.

m_{He} $\alpha \longrightarrow$ q_1

m_{Au} O Au. q_2 K.

金核多中. ⇒ 连续损失能量 ⇒ 当最近时. ⇒ O动量多.

⇒ O动量多中. 求解.

(1) 先找O动量多设. 此多. 速为 $v = \beta \cdot c$.

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

$$P_{地总} = P_{\alpha} = \sqrt{E_0^2/c^2 - m_{He}^2 c^2}$$

$$\beta = \frac{P_{地总}}{E_{地总}/c} \leftarrow$$

$$E_{地总} = E_0 + m_{Au} \cdot c^2.$$

$$\Rightarrow \beta = \frac{\sqrt{E_0^2 - m_{He}^2 c^4}}{E_0 + m_{Au} \cdot c^2}$$

$$\Rightarrow \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{E_0 + m_{Au} \cdot c^2}{\sqrt{m_{Au}^2 \cdot c^4 + m_{He}^2 c^4 + 2E_0 \cdot m_{Au} c^2 - m_{Au} c^2 - m_{He} c^2}}$$

⇒ O动量多中. 能量多值:

$$E_{O总} = m_{Au} \cdot c^2 + m_{He} \cdot c^2 + \frac{K q_1 q_2}{d}$$

$$E_{O总} = \gamma (E_{地总} - \beta \cdot P_{地总} \cdot c).$$

$$= \sqrt{m_{Au}^2 \cdot c^4 + m_{He}^2 \cdot c^4 + 2E_0 \cdot m_{Au} \cdot c^2}.$$

$$d = \frac{K \cdot q_1 \cdot q_2}{\sqrt{m_{Au}^2 c^4 + m_{He}^2 \cdot c^4 + 2E_0 \cdot m_{Au} \cdot c^2} - m_{Au} c^2 - m_{He} \cdot c^2}$$

题7.

水里有小气泡 研究大小

← 底片

← 镜头.

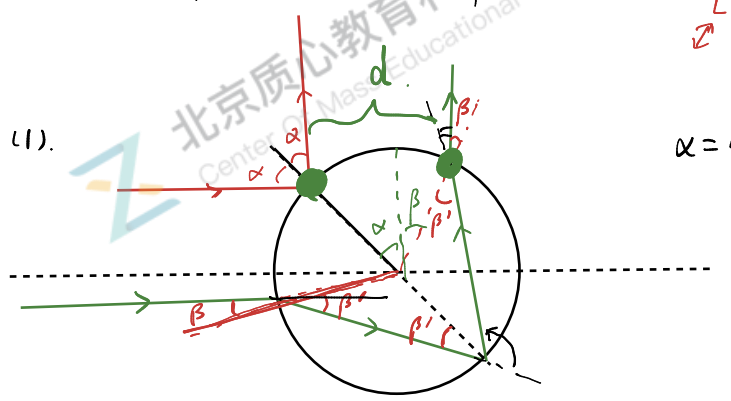
$n = 1.3$. $\lambda = 488\,nm$.

$f = 10\,cm$

$D = 3.6\,cm$

$L = 30\,cm$

(1).



$\alpha = 45°$ 即可

(2). 要求: $n \cdot \sin\beta = 1 \cdot \sin\beta'$.

$$\delta = \underbrace{\beta' - \beta}_{\text{顺}} - \underbrace{(\pi - 2\beta')}_{\text{逆}} + \underbrace{(\beta' - \beta)}_{\text{顺}} = -\frac{\pi}{2}$$

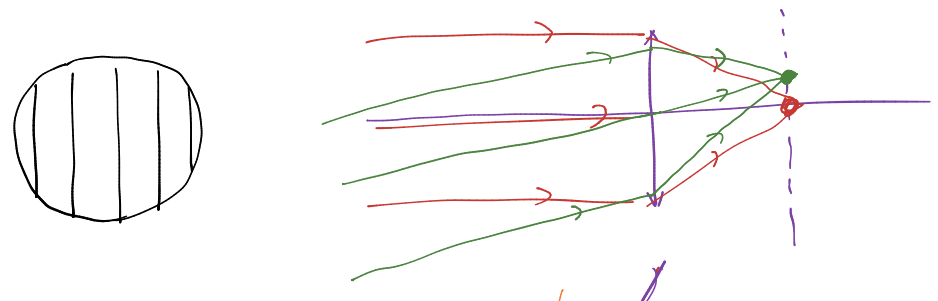
(逆)

⇒ $\beta = 0.468\,rad = 26.8°$.

不会怎么又: ① 底座不会刚成了吃干涉.

② 多次反射. 没有90°跳过一光.

(3). 透镜会出面成像: 经面位置 ⇔ 平行光角度



精确计算: 干涉:

$L = 30\,cm$.

气泡: $0.1mm$

$\Delta x = \frac{\lambda}{d} \cdot D$

$\Delta\theta = \frac{\Delta x}{D} = \frac{\lambda}{d}$

⇒ 只会角度细微一点点.

气泡内光程差一没变多.

⇒ 看干涉.

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$\Rightarrow d = r\cdot\sin\alpha + r\cdot\sin\beta = 1.158r$.

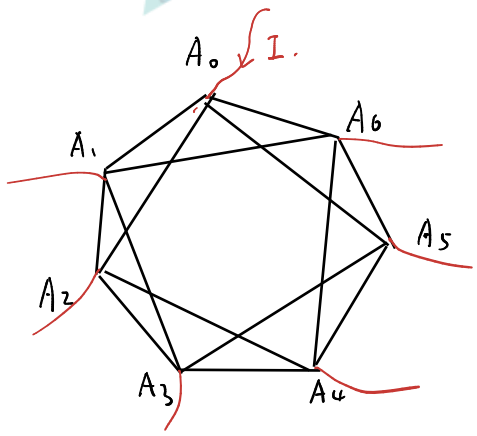
条纹角间距：$\Delta\theta = \frac{\lambda}{d}$

进入透镜之总角度为：$\theta = \frac{D}{L}$

$\Rightarrow$ 条纹数：$N = \frac{\theta}{\Delta\theta} = \frac{d\cdot D}{\lambda\cdot L} = \frac{1.158\cdot r\cdot D}{\lambda\cdot L}$

清晰：8～9条. $\Rightarrow$ $r = 28\mu m. \sim 32\mu m$.

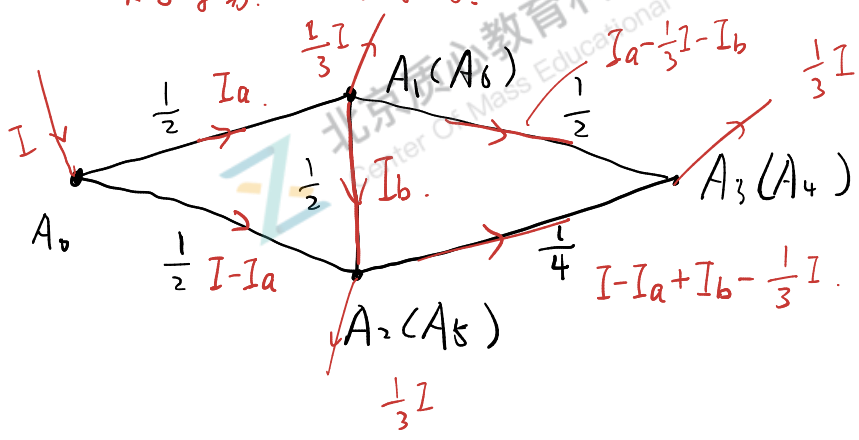
题8.



恢复对称性.

第1步：从A_0入I. 余下每个支路为 $\frac{I}{6}$.

1.6等势. 2.5等势. 3.4等势.



基二：左回路：$\frac{1}{2}I_a + \frac{1}{2}I_b - \frac{1}{2}(I - I_a) = 0$ ①

右回路：$\frac{1}{2}(I_a - \frac{1}{3}I - I_b) - \frac{1}{4}(I - I_a + I_b - \frac{1}{3}I)$

$-\frac{1}{2}I_b = 0$. ②

$\Rightarrow$ $I_a = \frac{19}{39}I$

$I_b = I - 2I_a = \frac{1}{39}I$.

$\Rightarrow$ $U_{01}^{(1)} = r\cdot\frac{I_a}{2} = \frac{19}{78}\cdot r\cdot I$.

(1) 第2步：从A_1出. I. 从余下每个支入 $\frac{I}{6}$.

1步+2步 $\Rightarrow$ A_0 入 $\frac{7}{6}I$. A_1 出 $\frac{7}{6}I$.

$R_{01} = \frac{2U_{01}^{(1)}}{\frac{7}{6}I} = \frac{6}{7}\times 2\times\frac{19}{78} = \frac{38}{91}r$.

(2). 2步：从A_2出 I. 从余下入 $\frac{I}{6}$.

$U_{02}^{(1)} = (I - I_a)\cdot\frac{r}{2}$

$= \frac{20}{39}I\cdot\frac{r}{2} = \frac{10}{39}Ir$

$\Rightarrow$ $R_{02} = \frac{2\cdot U_{02}^{(1)}}{\frac{7}{6}I} = \frac{6}{7}\times\frac{20}{39} = \frac{40}{91}r$

(3). R_{03}. 2步. 从A_3出 I. 从余下入 $\frac{I}{6}$.

$U_{03}^{(1)} = \frac{1}{2}rI_a + \frac{1}{2}r\cdot(I_a - \frac{1}{3}I - I_b)$

$= \frac{1}{2}\times\frac{19}{39}rI + \frac{1}{2}r\cdot(\frac{19-13-1}{39})$

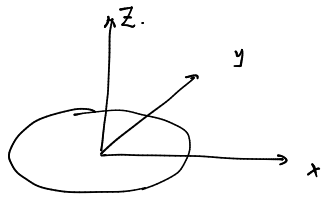
$= \frac{1}{2}\cdot(\frac{24}{39})rI$

$\Rightarrow$ $R_{03} = \frac{2\cdot U_{03}^{(1)}}{\frac{7}{6}I} = \frac{2}{7}\times\frac{24}{39} = \frac{48}{91}r$

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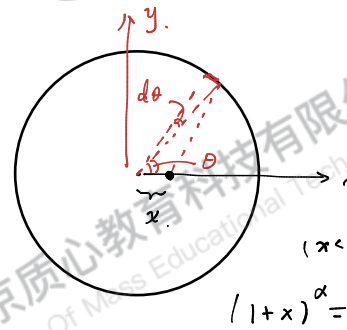
36套.

1. (1). 稳定.



对r小环.

$x \ll r$.



$q\cdot d\varphi = \dfrac{kq\cdot r\cdot d\theta\cdot\lambda}{\sqrt{x^2+r^2-2rx\cdot\cos\theta}}$

$(x<<1)$

$(1+x)^\alpha = 1+\alpha x+\dfrac{\alpha(\alpha-1)}{2!}x^2+\cdots$

$\Rightarrow q\cdot\varphi = \int_0^{2\pi}\dfrac{kq\cdot r\cdot\lambda\cdot d\theta}{\sqrt{x^2+r^2-2r\cdot x\cdot\cos\theta}}$

$= kq\cdot\lambda\cdot\int_0^{2\pi} d\theta\left(1+\dfrac{x^2}{r^2}-\dfrac{2x}{r}\cos\theta\right)^{-\frac{1}{2}}$

$= kq\lambda\int_0^{2\pi} d\theta\left(1-\dfrac{1}{2}\dfrac{x^2}{r^2}+\dfrac{x}{r}\cos\theta+\dfrac{-\frac{1}{2}(-\frac{3}{2})}{2!}\cdot\dfrac{4x^2}{r^2}\cdot\cos^2\theta\right)$

(常. 2π. $=0$. π.)

$= kq\cdot\lambda\cdot 2\pi\left(1+\dfrac{x^2}{4r^2}\right)$

$\Rightarrow q\cdot E_x = -\dfrac{d\varphi}{dx} = -\dfrac{kq\lambda\pi}{r^2}\cdot x$. 线性回复力.

对总共: $q\cdot E_{总x} = -\sum_{i=1}^{\infty}\dfrac{kq\lambda\pi}{(ir)^2}\cdot x$

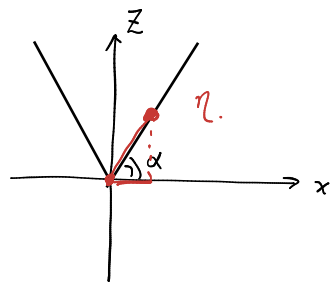
$= -\dfrac{kq\lambda\pi}{r^2}\cdot\dfrac{\pi^2}{6}\cdot x$.

$\Rightarrow$ 沿x振: $T = 2\pi\cdot\sqrt{\dfrac{6r^2\cdot m}{kq\cdot\lambda\pi^3}}$

(2) z方向: 受力沿+z方向 $\Rightarrow$ 不稳定.

(3). $x^2+y^2-kz^2=0$. $z=\dfrac{1}{\sqrt{k}}\cdot x$

$\tan\alpha = \dfrac{1}{\sqrt{k}}$.



由对称. 设小往标在xz平面内.

$\eta << r$.

$\Rightarrow$ 对r小环: $\varphi(\eta)$. $\Rightarrow E_\eta = -\dfrac{d\varphi}{d\eta}$.

$d\varphi = \dfrac{k\cdot\lambda\cdot r\cdot d\theta}{\sqrt{(r^2+(\eta\cdot\cos\alpha)^2-2r\cdot\eta\cdot\cos\alpha\cdot\cos\theta)+(\eta\cdot\sin\alpha)^2}}$

$= k\cdot\lambda\cdot d\theta\cdot\left(1+\dfrac{\eta^2}{r^2}-\dfrac{2\eta\cos\alpha}{r}\cdot\cos\theta\right)^{-\frac{1}{2}}$

$= k\cdot\lambda\cdot d\theta\left(1-\dfrac{1}{2}\cdot\dfrac{\eta^2}{r^2}+\dfrac{\eta\cdot\cos\alpha}{r}\cdot\cos\theta+\dfrac{-\frac{1}{2}(-\frac{3}{2})}{2}\cdot\dfrac{4\eta^2\cos^2\alpha}{r^2}\cdot\cos^2\theta\right)$

$(=2\pi)$

$\Rightarrow \varphi = \int_0^{2\pi} d\varphi = k\lambda\left(2\pi-\dfrac{\eta^2}{r^2}\cdot\pi+\dfrac{3}{2}\cdot\dfrac{\eta^2\cos^2\alpha}{r^2}\cdot\pi\right)$.

$\Rightarrow$ η方向电场:

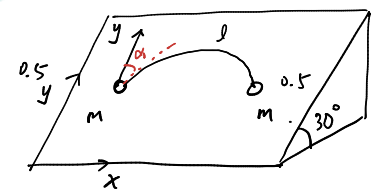
$E_{\eta(\eta)} = -\dfrac{d\varphi}{d\eta} = -k\lambda\left(-\dfrac{\pi}{r^2}+\dfrac{3}{2}\cdot\dfrac{\cos^2\alpha}{r^2}\cdot\pi\right)\cdot 2\eta$.

要稳定: $\dfrac{3}{2}\cdot\dfrac{\cos^2\alpha}{r^2}\pi-\dfrac{\pi}{r^2}>0$

$\Rightarrow \cos^2\alpha > \dfrac{2}{3}$. $\tan\alpha = \dfrac{1}{\sqrt{k}}$

$\cos^2\alpha = \dfrac{1}{1+\tan^2\alpha} = \dfrac{1}{1+\frac{1}{k}} > \dfrac{2}{3}$. $\Rightarrow k>2$. $\Rightarrow$ ✓.

题2.

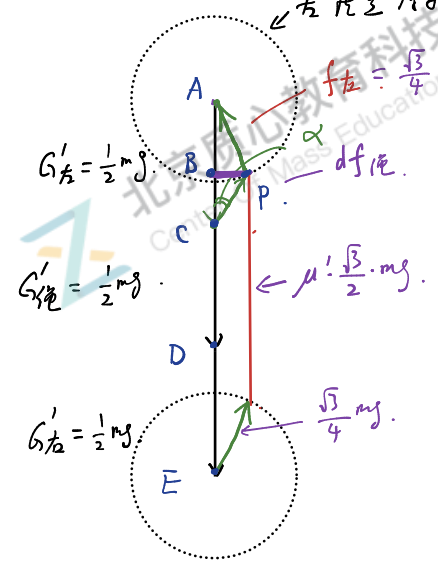


$\sqrt{3}-1 < \mu' < \dfrac{\sqrt{3}}{2}$

平面内: $g_1 = g\cdot\sin 30° = \dfrac{1}{2}g$. ← 平面内等效重力加速度

$f_{球max} = \mu\cdot mg\cdot\cos 30° = \dfrac{1}{2}\cdot mg\cdot\dfrac{\sqrt{3}}{2} = \dfrac{\sqrt{3}}{4}mg$.

平面内质点受力:



$\sum|df_{地}| = \mu'\cdot mg\cos\theta$

$f_{地}$. 总长固定

若总长够长则:

相切时α最大.

若总长不够长则. f地连接处外处α最大.

由题给条件. 可以平衡. 但α不能到切线.

$\Rightarrow AB = \dfrac{1}{2}\cdot\left(\dfrac{3}{2}mg-\mu'\cdot\dfrac{\sqrt{3}}{2}mg\right)$

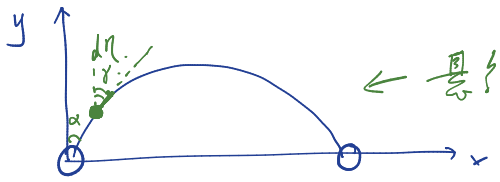
$= \left(\dfrac{3}{4}-\dfrac{\sqrt{3}}{4}\mu'\right)mg$

$\Rightarrow BP = \sqrt{AP^2-AB^2}$, $BC = \dfrac{1}{2}-AB = \cdots$

$\Rightarrow \tan\alpha = \dfrac{BP}{BC} = \cdots = \dfrac{\sqrt{3-(3-\sqrt{3}\mu')^2}}{\sqrt{3}\mu'-1}$

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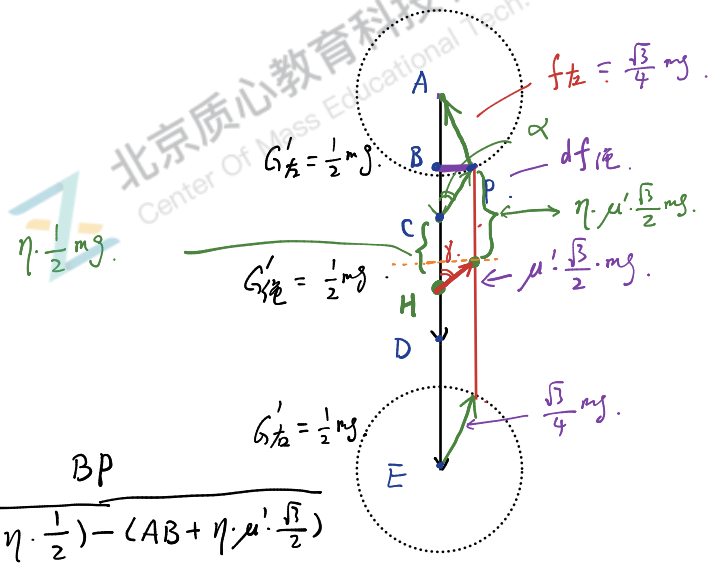
(2).



xy:

$dx = d\eta \cdot \sin\gamma$. $(0 \le \eta < \frac{1}{2})$.

取. 左起点. 及. 右边. $\eta \cdot l$ 长. 绳子. 为研究对象.

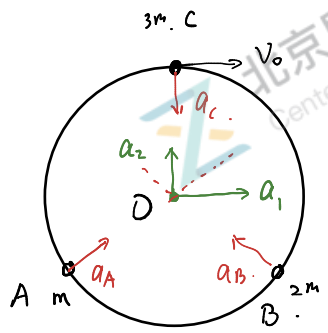


$$\Rightarrow \tan\gamma = \frac{BP}{(\frac{1}{2} + \eta\cdot\frac{1}{2}) - (AB + \eta\cdot\mu'\cdot\frac{\sqrt{3}}{2})}$$

$$= \frac{\sqrt{(\frac{\sqrt{3}}{4})^2 - (\frac{1}{2} - (\frac{\sqrt{3}}{2}\mu' - \frac{1}{2})/2)^2}}{\frac{1}{2}(\frac{\sqrt{3}}{2}\mu' - \frac{1}{2})}$$

$$\Rightarrow L = 2\int_0^{\frac{1}{2}} l\cdot\sin\gamma\cdot d\eta \xrightarrow{\text{计算机积}} 0.896l.$$

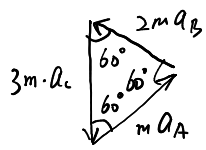
题3.



光滑. ⇒ 受力只能沿半径 ⇒ 加速度只能沿半径.

设 a_A. a_B. a_C

对整体. 合外力0:



设 $a_A = a$, $a_B = \frac{1}{2}a$. $a_C = \frac{1}{3}a$. 方向如图.

只有C有速度. 由以下: $v_0 = 0$, v_A, $v_B = 0$

⇒ AO以下: 相对加速度为0. ⇒ $\frac{1}{2}a_2 + \frac{\sqrt{3}}{2}a_1 = a$ ①

BO以下: 相对加速度为0. ⇒ $\frac{1}{2}a_2 - \frac{\sqrt{3}}{2}a_1 = \frac{1}{2}a$. ②

CO以下: $a_{C0} = a_{向心} = \frac{v_0^2}{R}$.

$$\frac{1}{3}a + a_2 = \frac{v_0^2}{R}.$$

$$\Rightarrow a_A = \frac{6}{11}\frac{v_0^2}{R},\ a_B = \frac{3}{11}\frac{v_0^2}{R},\ a_C = \frac{2}{11}\frac{v_0^2}{R}.$$

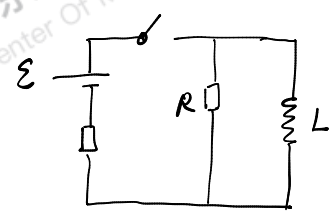
方向如图.

$$a_1 = \frac{3}{2}\cdot a_A = \frac{9}{11}\frac{v_0^2}{R}$$

$$a_2 = \frac{1}{2\sqrt{3}}a_A = \frac{\sqrt{3}}{11}\cdot\frac{v_0^2}{R}$$

$$\Rightarrow a_O = \frac{2\sqrt{21}}{11}\cdot\frac{v_0^2}{R}.$$

题4.



L:

闭合期间: q_0

断开: Q_0.

(1). 能量: $Q_0 = \frac{1}{2}L\cdot I_0^2 \Rightarrow I_0 = \sqrt{\frac{2Q_0}{L}}$

(2) 其二: $R\cdot i_R = L\cdot\frac{di_L}{dt}$

$$\int_0^{q_R} R\cdot dq_R = \int_0^{I_0} L\cdot di_L.$$

$$\Rightarrow q_R = \frac{LI_0}{R} = \frac{1}{R}\sqrt{2LQ_0}$$

(3) 同理: $q_2 = \frac{\sqrt{2LQ_0}}{R}$. (反向流动).

(4). 流过电源总电荷: $q_0 + q_1$.

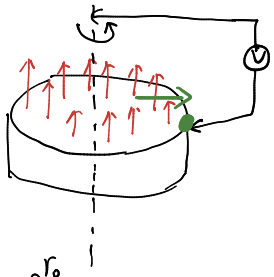
电源做功: $W = \varepsilon\cdot(q_0 + q_1) = \cdots$

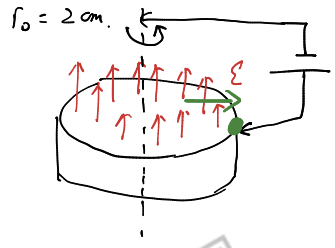
(5). 能量守恒. $W - \frac{1}{2}LI_0^2 = Q_{热}$.

$$\Rightarrow Q_{热} = W - Q_0 = \cdots$$

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题5.





(1) $V = -\int_0^{r_0} B \cdot \omega \cdot r \cdot dr = -\frac{1}{2}\omega r_0^2 \cdot B = -63\,mV$.

(2) 最大速：$\frac{1}{2}\omega \cdot r_0^2 \cdot B = \mathcal{E} \Rightarrow$ 转速：7.2×10^4 转/分.

逆时针.

(3) $\omega = 0$ 时，运动力矩最 $\Rightarrow \beta$ 最大.

$\Rightarrow I\beta_{max} = B \cdot r_0 \cdot \frac{r_0}{2} \cdot \frac{\mathcal{E}}{R}$

$\Rightarrow \beta_{max} = \frac{\mathcal{E}B}{mR}$

最大角速：力矩平衡时.

$k \cdot \omega_{max} = I \cdot B \cdot \frac{r_0^2}{2} = \frac{\left(\mathcal{E} - \frac{1}{2}\cdot\omega_{max}\cdot r_0^2 \cdot B\right)}{R} \cdot B \cdot \frac{r_0^2}{2}$

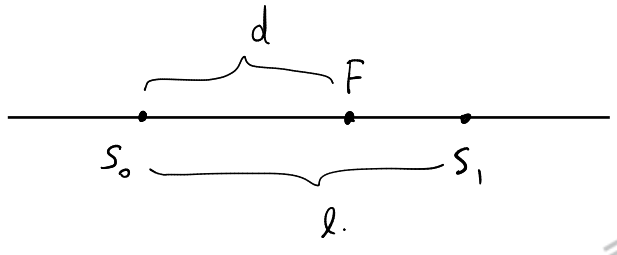
$\Rightarrow \omega_{max} = \frac{2\mathcal{E}\cdot B\cdot r_0^2}{4kR + B^2 r_0^4}$

(4) $M_0 + k\omega = \frac{\left(\mathcal{E} - \frac{1}{2}\cdot\omega_{max}\cdot r_0^2\cdot B\right)}{R} \cdot B \cdot \frac{r_0^2}{2}$

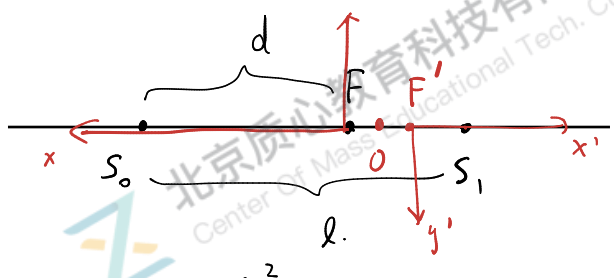
$P = M_0 \cdot \omega = \cdots = \frac{2\mathcal{E}Br_0^2 M_0 - 4RM_0^2}{4kR + B^2 r_0^4}$

$\Rightarrow$ 当 $M_0 = \frac{\mathcal{E}B\cdot r_0^2}{4R}$ 时负载功率最大.

题6.



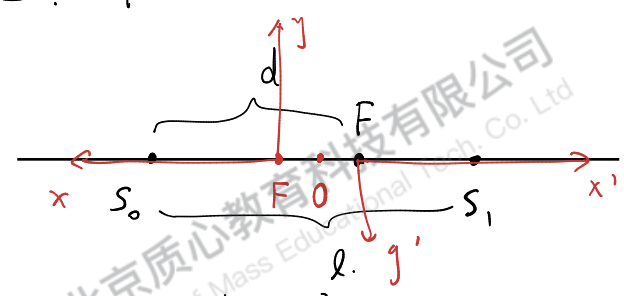
(1). 情况一. F 是物方焦点.



$d\cdot(\ell - d - 2f) = f^2$.

$\Rightarrow f = -d \pm \sqrt{d\ell}$.

情况二：F 是像方焦点.

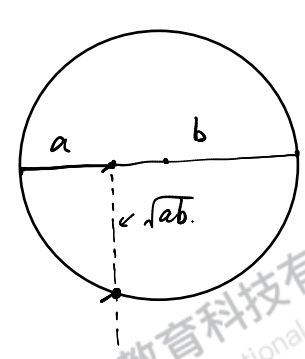


$(d-2f)(\ell-d) = f^2$

$\Rightarrow f = -(\ell - d) \pm \sqrt{(\ell-d)\cdot\ell}$.

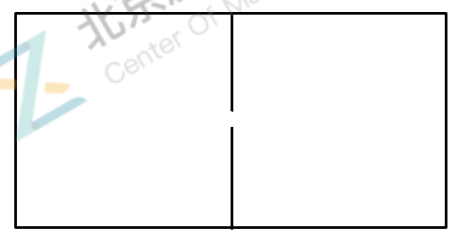
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(2). 只需要会画 $\sqrt{ab}$ 即可.



$\Rightarrow$ ✓

题7.



$a = 1.5R$.

初：T_0, V_0.

左：$2p_0$. 右 p_0.

(1). 左：状态：$p\cdot V_0 = n_L \cdot RT$. $\Rightarrow n_L = \frac{pV_0}{RT}$

$n_R = n_{总} - n_L = \frac{3p_0V_0}{RT_0} - \frac{pV_0}{RT}$.

(2) $pV^\gamma = 常 \Rightarrow p^{1-\gamma}\cdot T^\gamma = 常$. $\gamma = \frac{5}{3}$

$\Rightarrow$ 左：$p^{1-\gamma}\cdot T^\gamma = (2p_0)^{1-\gamma}\cdot T_0^\gamma$

$\Rightarrow T = T_0\cdot\left(\frac{2p_0}{p}\right)^{\frac{1-\gamma}{\gamma}} = T_0\cdot\left(\frac{p}{2p_0}\right)^{\frac{2}{5}}$

(3) 能：$n_L\cdot C_v\cdot T_L + n_R\cdot C_v\cdot T_R = n_{总}\cdot C_v\cdot T_0$.

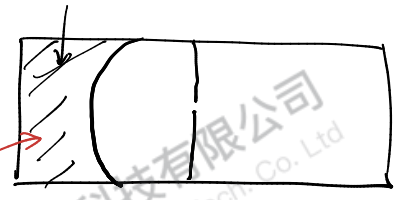
用 $C_v = \frac{3}{2}R$, $PV = \nu RT$.

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$$\frac{3}{2}\cdot P_f\cdot V_0+\frac{3}{2}P_f\cdot V_0=\frac{3}{2}\cdot 3P_0\cdot V_0 \quad ①$$

$$\Rightarrow P_f=\frac{3}{2}P_0$$

n_L. 最后还在充气的气体



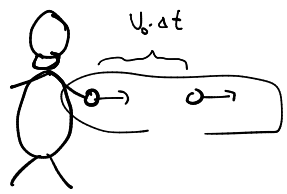
$$\left(\frac{n_L\cdot V_0}{\frac{2}{3}\cdot(n_L+n_R)}\right)$$

$\Rightarrow$ n_L. 最后还在充气的气体：绝热方程：

$$2P_0\cdot\left(\frac{n_L}{\frac{2}{3}(n_L+n_R)}\cdot V_0\right)^{\gamma}=P_f\cdot V_0^{\gamma}.$$

$$\Rightarrow \quad \chi=\cdots=1.28$$

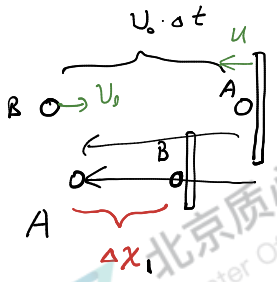
题8.





$$V_0=u=\frac{c}{2}.$$

(1).



设球反射回去速度为：V'

循环中 每隔 $\Delta t_1=\frac{V_0\cdot\Delta t}{V_0+u}$ 打到一个球

↘ 回来2球间距：

$$\Delta x_1=V'\Delta t_1-u\cdot\Delta t_1=\frac{V_0(V'-u)}{V_0+u}\cdot\Delta t$$

求 V'：

先. 球打之前. 球相对谁的速度：

$$V_1=\frac{V_0+u}{1+\frac{V_0\cdot u}{c^2}}$$

V_1 ← ← u_0

再：打回之后. 换回循环.

$$\Rightarrow |V'|=\frac{V_1+u_0}{1+\frac{V_1\cdot u_0}{c^2}}=\frac{V_0+2u+\frac{V_0u^2}{c^2}}{1+\frac{V_0u}{c^2}+\frac{V_0u}{c^2}+\frac{u^2}{c^2}}=\frac{13}{14}c.$$

循环前：$\Delta t_2=\frac{\Delta x_1}{V'}=\cdots$

$$\Rightarrow \Delta t_1=\cdots=\frac{3}{13}\cdot\Delta t.$$

(2) 循环中：$F=\frac{\Delta P_{动}}{\Delta t_1}$

$$=\frac{\frac{1}{\sqrt{1-\frac{1}{2}^2}}\cdot m_0\cdot\frac{1}{2}c+\frac{1}{\sqrt{1-V'^2/c^2}}\cdot m_0\cdot V'}{\Delta t_1}$$

$$=\frac{32\sqrt{3}}{9}\cdot\frac{mc}{\Delta t}.$$

(3). $P_{功率}=F\cdot u=\frac{16\sqrt{3}}{9}\cdot\frac{mc^2}{\Delta t}\Rightarrow\surd$.

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